

Matrix functions and stability

NP-hardness of unrestricted static output-feedback stabilization

Difficulty: extreme **Importance:** broadly interesting

Status: Solved

Historical rating rationale: Extreme because this is a longstanding complexity barrier for unrestricted feedback design; broad impact connects control synthesis, optimization, and computational complexity.

Resolution — 18 September 2026

Affirmative resolution. Unrestricted static output-feedback stabilization is NP-hard. Johan Löfberg proves this for integer single-input plants in [Theorem 1.1 of arXiv:2609.16886v1](#). Amir Ali Ahmadi, Abraar Chaudhry, Ijay Narang and Yukai Tang prove strong NP-hardness for integer single-output plants in [Theorem 1 of arXiv:2609.20636v1](#).

Each integer subclass is contained in the original rational-input language: encode integers with denominator one, retaining the same unrestricted real gain and the strict Hurwitz condition. Composing this inclusion with either paper's polynomial-time many-one reduction settles the retained question. The theorem credit belongs to the cited authors.

Review level. These are v1 preprints submitted on 15 and 17 September 2026. A separate Codex-agent review examined both complete continuous-time arguments and their match to this problem; no substantive gap was found. This is informal review, with no claim of journal acceptance, external human peer review or Lean verification. [Reference note and review scope](#).

Literature application and curation: George Stepaniants, Department of Computing and Mathematical Sciences, California Institute of Technology. The difficulty and importance ratings above are historical. The original statement and the 10 September audit are retained.

Problem statement

Consider the decision language whose inputs are rational matrices $A \in \mathbb{Q}^{n \times n}$, $B \in \mathbb{Q}^{n \times m}$, and $C \in \mathbb{Q}^{p \times n}$, with positive dimensions included in the input. The answer is yes precisely when there exists an unrestricted real matrix $K \in \mathbb{R}^{m \times p}$ for which

$$\operatorname{Re} \lambda < 0 \quad \text{for every } \lambda \in \sigma(A + BKC).$$

Is this language NP-hard under polynomial-time many-one reductions in the binary encoding of the rational input? The requested output is a complexity theorem, not another numerical heuristic for finding K .

References and status evidence

Minyue Fu, [Two Challenging Problems in Control Theory](#), §3, pp. 3–7, defines strict Hurwitz stabilization and distinguishes it from the known NP-hard problem with entrywise bounds on K (§3.1). Gillis and Sharma, [Solving matrix nearness problems via Hamiltonian systems, matrix factorization, and optimization](#), §3.5.2, pp. 50–51 (2022 manuscript), reiterates the unresolved complexity of static output feedback. Its broader stability convention should not replace the strict inequality above. Follow-up searches for unrestricted stabilization NP-hardness found no resolving reduction; claims about arbitrary bilinear matrix inequalities or prescribed pole placement do not by themselves settle this language. No claim of NP membership is made.

Audit — 2026-09-10

Rechecked [Fu, §3.1](#) and [Gillis–Sharma, §3.5.2](#). Searches for unrestricted static-output-feedback NP-hardness found no applicable reduction. Bounded feedback gains, prescribed poles, and general bilinear inequalities remain different decision problems; later generic hardness descriptions do not establish this one.